A

Mini Project Report

on

Blood Donation App (Blooddrop)

Submitted in partial fulfillment of the requirements for the degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Blood Donation App (Blooddrop)** has been submitted by **Bhavesh Bonde_(24107046)**, **Hansika Gondhale_(24107062)** , **Pranjali Gujar_(24107119)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Prof. Kiran Gaikwad**. Expressing gratitude towards our HOD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Hardiki Patil and Prof Dipali Gatt** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1:

Introduction

The Blood Donation Management System is an Android-based application developed to connect blood donors with recipients in an efficient and organized manner. In emergency situations, finding the right blood donor quickly is very important. This system helps reduce the time and effort required to locate suitable donors.

The application provides a platform where users can register, log in, and perform actions based on their roles such as donor, recipient, or admin. Donors can view blood requests and respond, while recipients can raise requests for required blood groups.

1.1 Purpose

- To design and develop a system for blood donation and request management
- To simplify the process of finding and donating blood
- To provide a user-friendly interface for users
- To ensure proper coordination between donors and recipients

1.2 Problem Statement

In many cases, patients face difficulty in finding blood donors during emergencies. Traditional methods such as phone calls and manual searching are slow and inefficient. There is a need for a digital system that can manage donor information and provide quick access to blood availability.

1.3 Objectives

1.To enable blood search and matching functionality:

The system allows users to search for required blood groups and quickly find suitable donors, helping reduce time during emergency situations.

2.To implement authentication and role management:

The application provides secure login and assigns roles (donor, recipient, admin) so that each user can access only the features relevant to them.

3.To ensure data validation and privacy:

Proper input validation is applied to avoid incorrect data entry, and secure handling techniques are used to protect user information.

4.To develop an admin module:

The admin panel helps in managing donor details, blood requests, and overall system activities efficiently, ensuring smooth operation of the application.

1.4 Scope

- Allows users to register and log in securely to access the application
- Supports basic blood group matching to connect donors and recipients
- Enables creation and management of blood requests
- Focuses on efficient handling of user data and request information
- Provides a simple and organized system for managing blood donation activities.

Chapter 2

Proposed System

The proposed system is designed to provide an efficient platform for managing blood donation activities. It allows users to interact with the system through a mobile application.

The system identifies users based on their roles and provides different functionalities accordingly. Donors can view requests and respond, while recipients can raise requests. The admin monitors and manages all activities within the system.

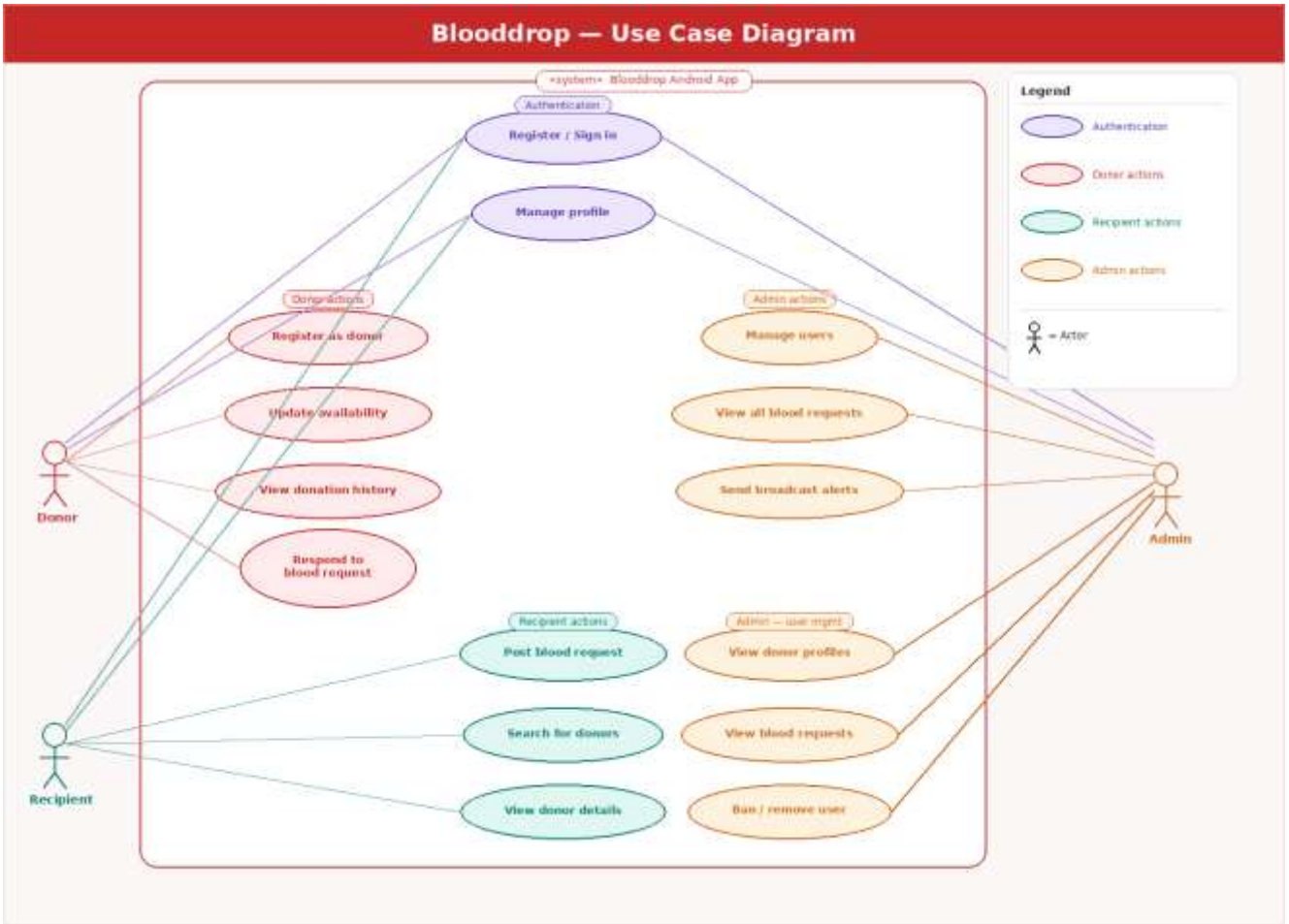

Figure 2.1 Use Case Diagram

2.1 Features and Functionality

1.User authentication and role-based access:

The system allows users to securely register and log in. Based on their role (donor, recipient, or admin), they are given access only to the features relevant to them, ensuring security and proper control.

2.Blood request creation and management:

Recipients can create blood requests by specifying details like blood group and requirements. The system stores and manages these requests, allowing donors to view and respond accordingly.

3.Basic blood group matching system:

The application matches donors and recipients based on compatible blood groups, helping users quickly find suitable donors for a request.

4.Admin management system:

The admin panel enables monitoring and control of the system, including managing users, verifying donors, and handling blood requests efficiently.

Chapter 3
Project Outcomes

The project demonstrates the ability to design and develop a functional Android-based application that addresses real-world problems related to blood donation. It integrates a user-friendly interface with backend services to ensure smooth and efficient operation. The system effectively connects donors and recipients, improving accessibility and reducing the time required to find suitable donors during emergencies. It also highlights the practical application of software development concepts in solving real-life issues.

The system showcases the implementation of secure authentication and role-based access control. Different user roles such as donor, recipient, and admin are managed efficiently, ensuring that each user can access only the features relevant to them. This improves system security, maintains proper control, and ensures organized handling of user activities within the application.

The application successfully implements core functionalities such as blood request creation, management, and blood group matching. It allows recipients to raise requests and donors to respond accordingly, making the process simple and efficient. The matching system helps in quickly identifying suitable donors, thereby improving the effectiveness of the blood donation process.

The project also reflects effective data management and system organization through the use of Firebase services. It ensures proper storage, retrieval, and handling of user and request data. The admin module enhances system control by allowing monitoring and management of users and activities. Overall, the system provides a reliable and structured approach to managing blood donation processes.

Chapter 4
Software Requirement

Operating System

- Android OS (for mobile deployment)
- Windows (for development environment)

Development Tools

- Android Studio (for app development)
- Android SDK (for building and testing the application)

Programming Language

- Java and Kotlin (for application development and logic implementation)

Front End (GUI)

- The application is developed using Android Studio
- XML is used to design user interface screens like Login, Registration, Role Selection, and Blood Request
- Java/Kotlin is used to handle user interactions and application logic

Back End (Database)

- Firebase is used as the backend platform
- Realtime Database is used to store user data such as name, blood group, location, and contact details
- Firebase Authentication is used for secure login and role-based access control

Libraries & Services Used

- Firebase Authentication → for user login and security
- Realtime Database → for storing and managing data

Chapter 5
Project Design

Architecture Components

1. Frontend (Mobile Application)

- Built using **Android (Kotlin)** with Android Studio
- Provides UI for:
  - Donor registration & availability
  - Blood request posting
  - Donor search & profile view
  - Admin dashboard (manage users, requests, alerts)

2. Backend (Firebase Services)

- **Firebase Authentication** → User login / signup (Donor, Recipient, Admin)
- **Cloud Firestore** → Stores:
  - User profiles (name, blood group, location)
  - Blood requests
  - Admin actions / ban records

3. System Modules

- **User Authentication Module** → Login/signup via Firebase
- **Donor Management Module** → Register, update availability, view history
- **Blood Request Module** → Post requests, match with available donors
- **Search & Match Module** → Filter donors by blood group and location
- **Admin Control Module** → Manage users, view all requests, send alerts, ban users

DATAFLOW

Donor / Recipient / Admin → Android App (Kotlin) → Firebase Authentication →stores profiles, requests, history, availability, admin actions → Search Module (blood group) → Firebase Cloud Messaging → Donor alerted + Recipient → Donation recorded in Firestore

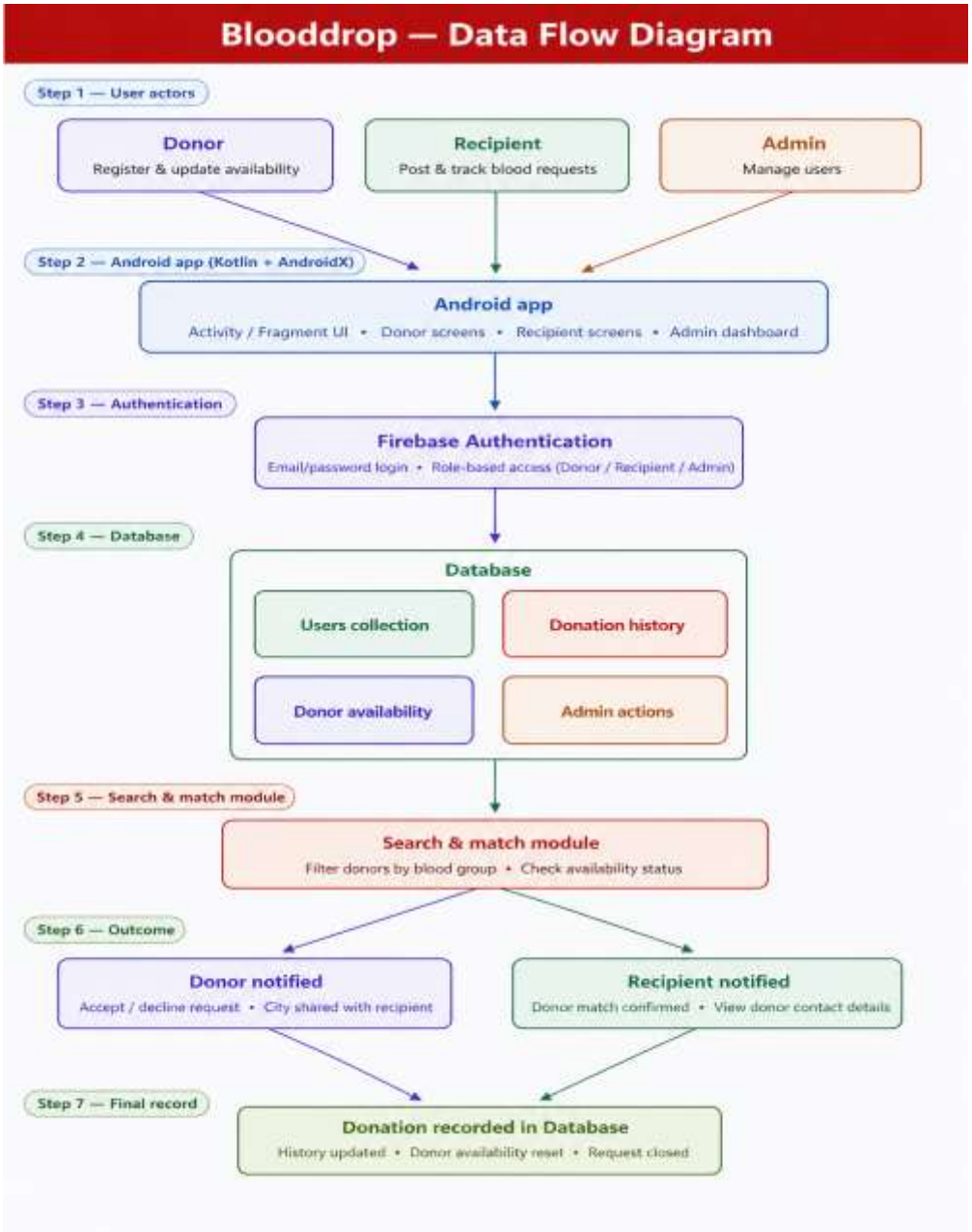

Figure 5.1:Data Flow Diagram

Chapter 6

Project Scheduling

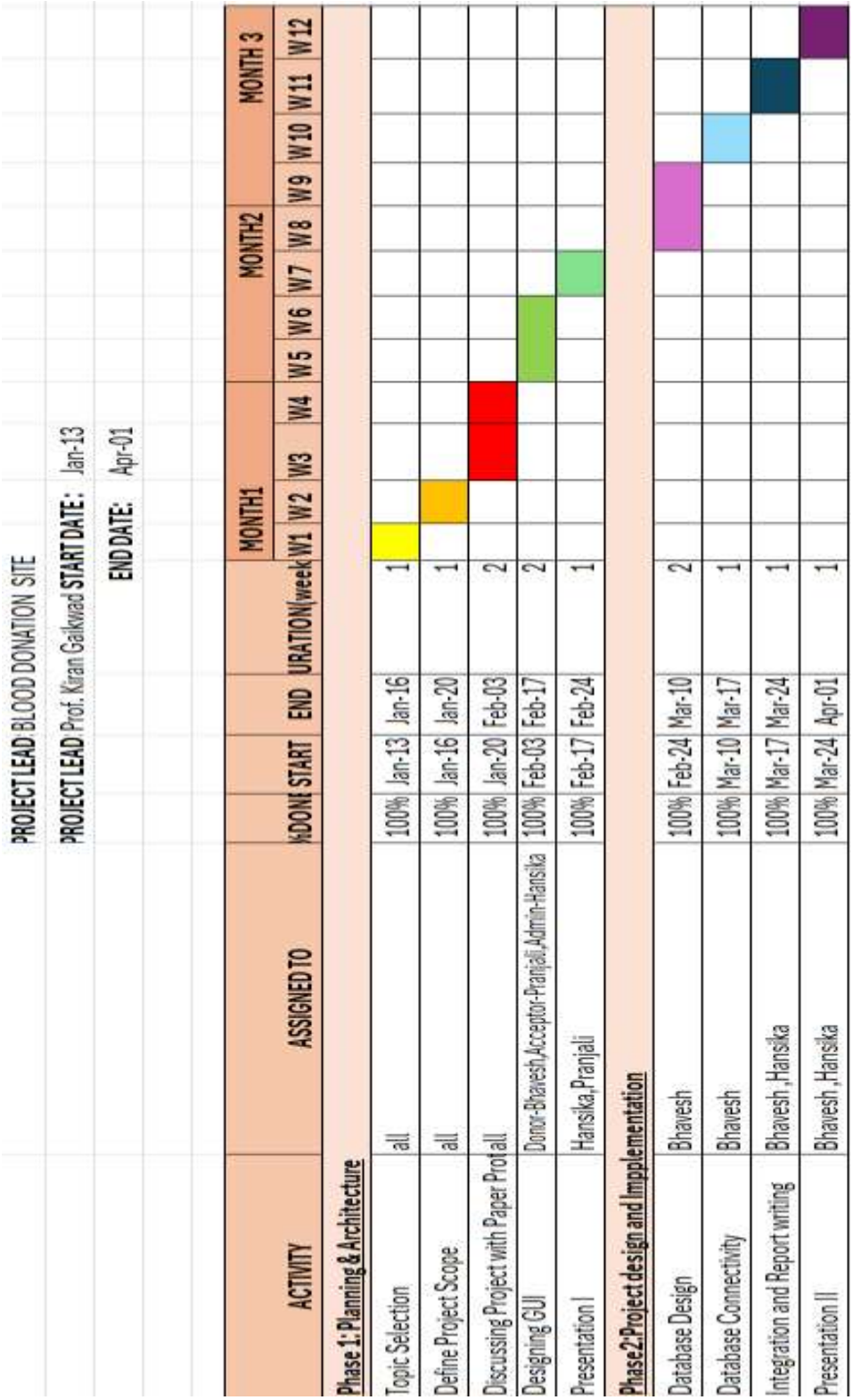

Figure 6.1: Gantt Chart for Project Scheduling

Chapter 7
Results

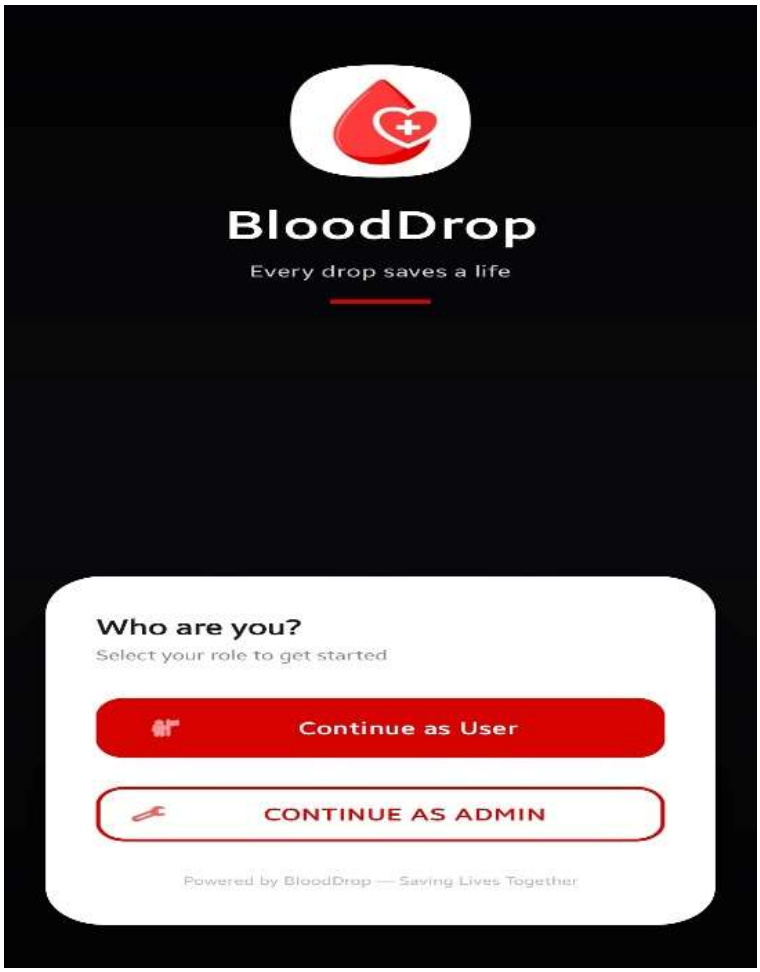

Figure 7.1:This is the role selection screen of the BloodDrop app

This screen represents the welcome and role selection page of the BloodDrop application, serving as the entry point for users. It displays the app logo and tagline, creating a clear identity and purpose. The page prompts users to choose their role allowing them to continue either as a normal user (donor/recipient) or as an admin. This helps in directing users to the appropriate functionality based on their role. The interface is simple, clean, and user-friendly, ensuring easy navigation. Overall, this page plays an important role in enabling role-based access and guiding users into the system efficiently.

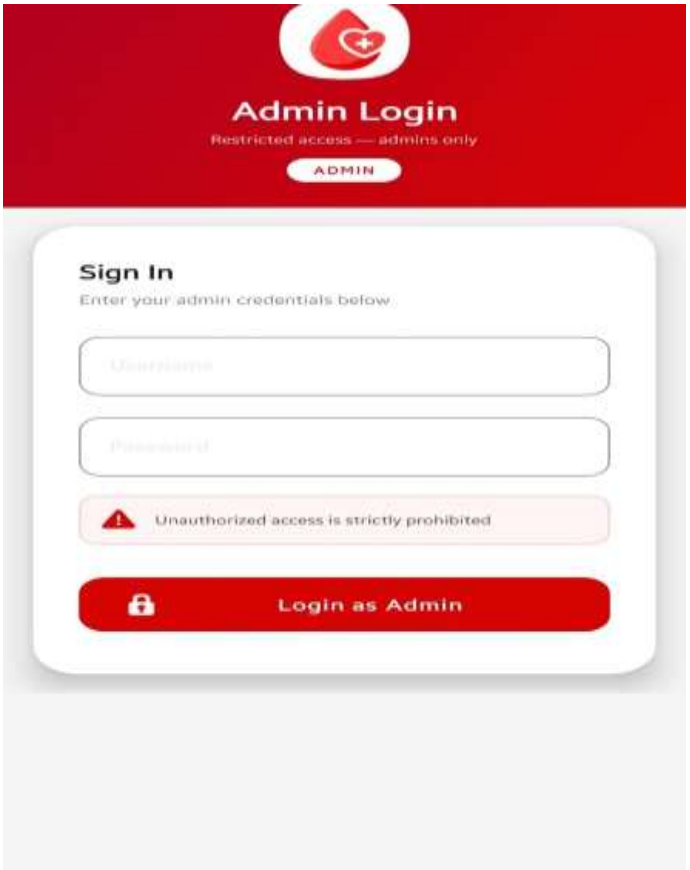

Figure 7.2: Admin Login page

This screen represents the Admin Login page of the BloodDrop application, designed specifically for restricted access. It allows only authorized administrators to log in using valid credentials such as username and password. The page clearly indicates that it is meant for admins only, ensuring security and controlled access. A warning message is displayed to prevent unauthorized usage. The interface is simple and focused, making it easy for admins to enter their details. Once logged in, the admin can manage users, blood requests, and overall system activities. This page plays a crucial role in maintaining system control and data security.

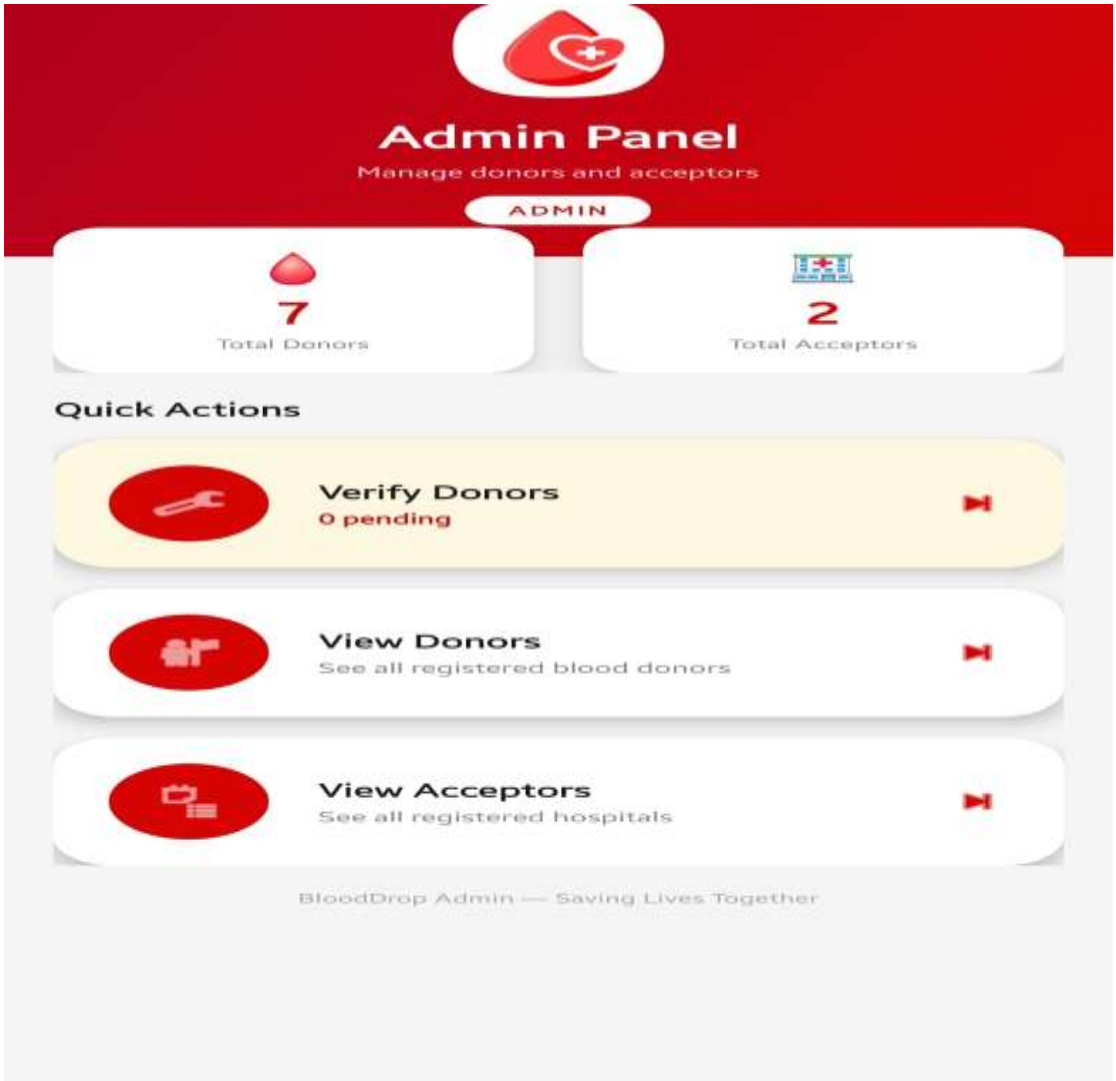

Figure7.3: Admin Panel dashboard

This screen represents the Admin Panel dashboard of the BloodDrop application, where the admin can monitor and manage the entire system. It displays key statistics such as the total number of donors and acceptors registered in the system. The “Quick Actions” section provides options like verifying donors, viewing donor details, and viewing acceptor (hospital) information. The admin can ensure that donor data is valid and up to date through the verification feature. This panel allows efficient control over user records and activities. Overall, it helps maintain the system’s reliability, organization, and proper functioning.

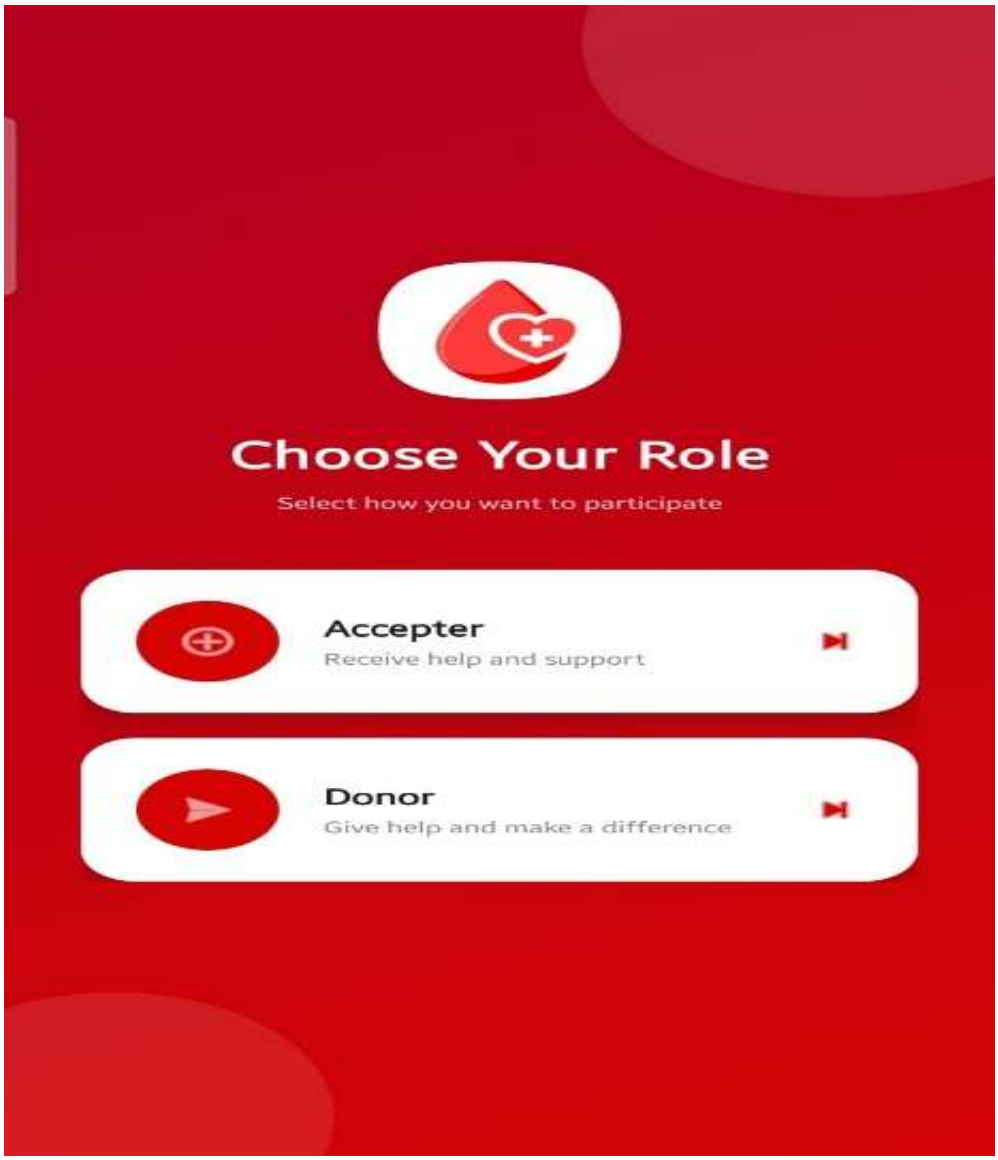

Figure 7.4: choose their role as either a donor or an accepter (recipient)

This screen is a role selection page that asks you to choose how you want to use the app. It features a clean red design with the title “Choose Your Role” and a short instruction to select your participation type. There are two main options: Accepter, which is for users who want to receive help and support, and Donor, which is for users who want to give help and make a difference. Each option is presented as a button that you can tap to proceed. Your selection determines how the app will be customized to match your needs and actions moving forward.

Figure7.5:This screen shows the Acceptor (hospital) registration page

This screen is the Accepter Registration page where users who need help can enter their details. It asks for hospital-related information such as the hospital name, specialty, address, and location. Below that, there is a section for contact details, including the contact person's name, phone number, and email. The layout is simple and organized, making it easy to fill in the required fields. This information helps the app understand the user's needs and connect them with the right support. Once completed, the user can proceed to register and start receiving assistance.

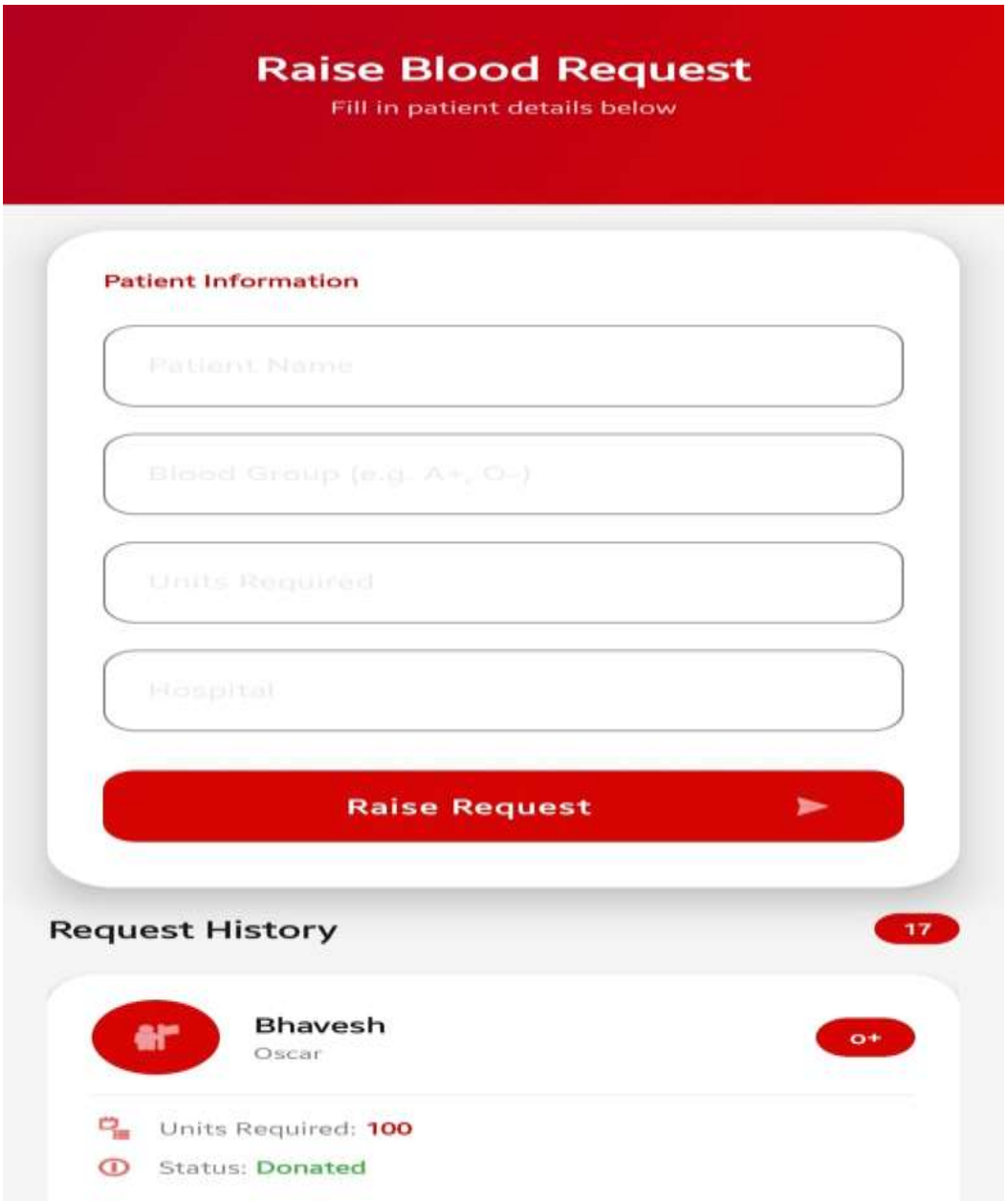

Figure7.6: users to submit new blood requests via a patient detail form

This screen is used to raise a blood request by entering patient details. At the top, it shows the title "Raise Blood Request" with instructions to fill in the required information. The form includes fields like patient name, blood group, units required, and hospital name. After filling in the details, the user can tap the "Raise Request" button to submit the request. Below the form, there is a Request History section that displays previous requests, including patient names, blood groups, required units, and donation status. This helps users track past requests and see whether they have been fulfilled.

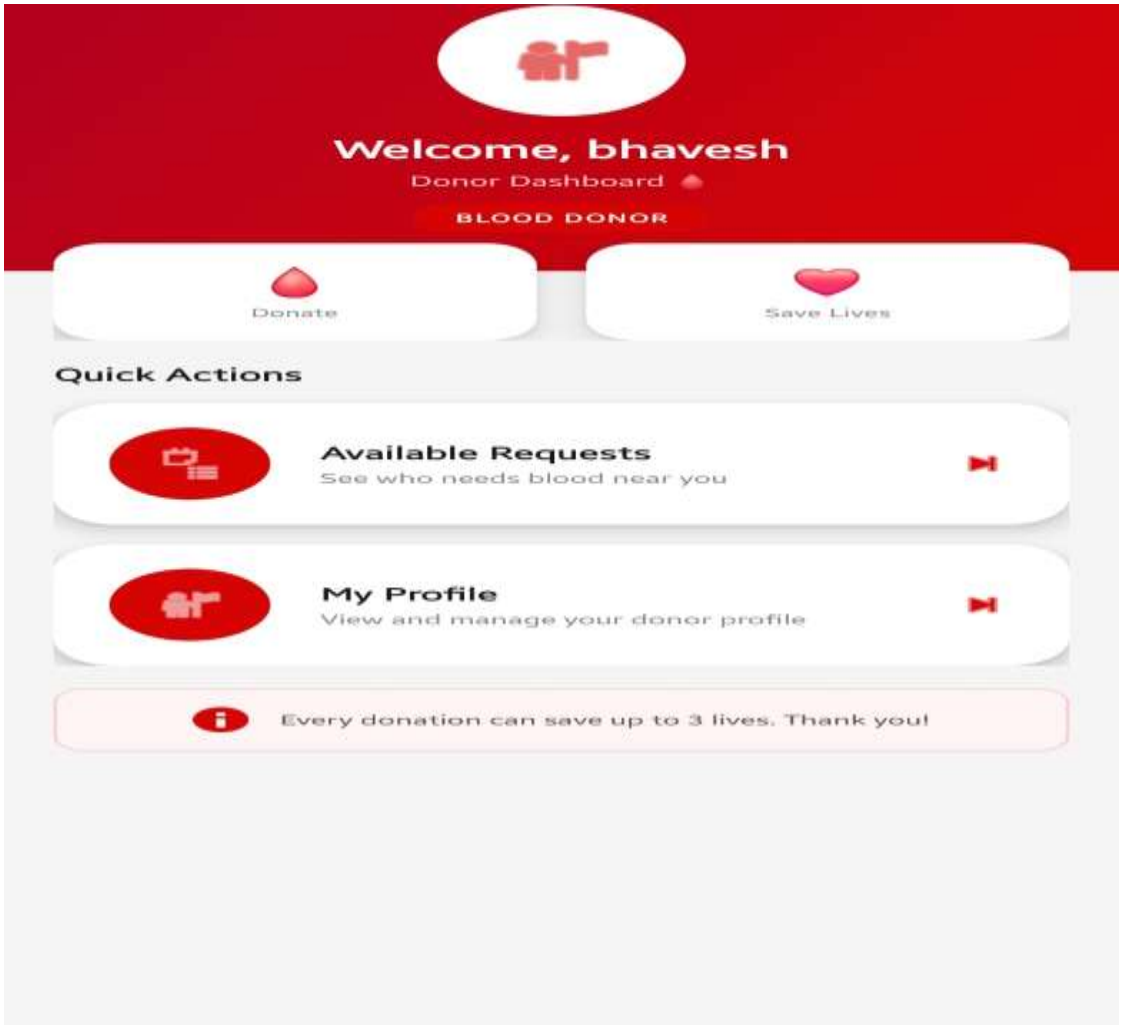

Figure7.7:This donor dashboard

This blood donor dashboard features a welcoming interface with a red theme, greeting the user by name and clearly labeling the section for donors. Prominently displayed at the top are two main action buttons, "Donate" and "Save Lives," utilizing blood drop and heart icons to encourage immediate engagement. Below these, the "Quick Actions" section provides easy access to "Available Requests" for finding local needs and "My Profile" for personal management. The layout is designed for efficiency, allowing donors to quickly navigate essential functions without confusion. At the bottom, an inspiring message reminds users that every donation can save up to three lives, adding a motivational touch to the user experience. Overall, the dashboard effectively combines intuitive navigation with a call to service, focusing on connecting donors with those in need.

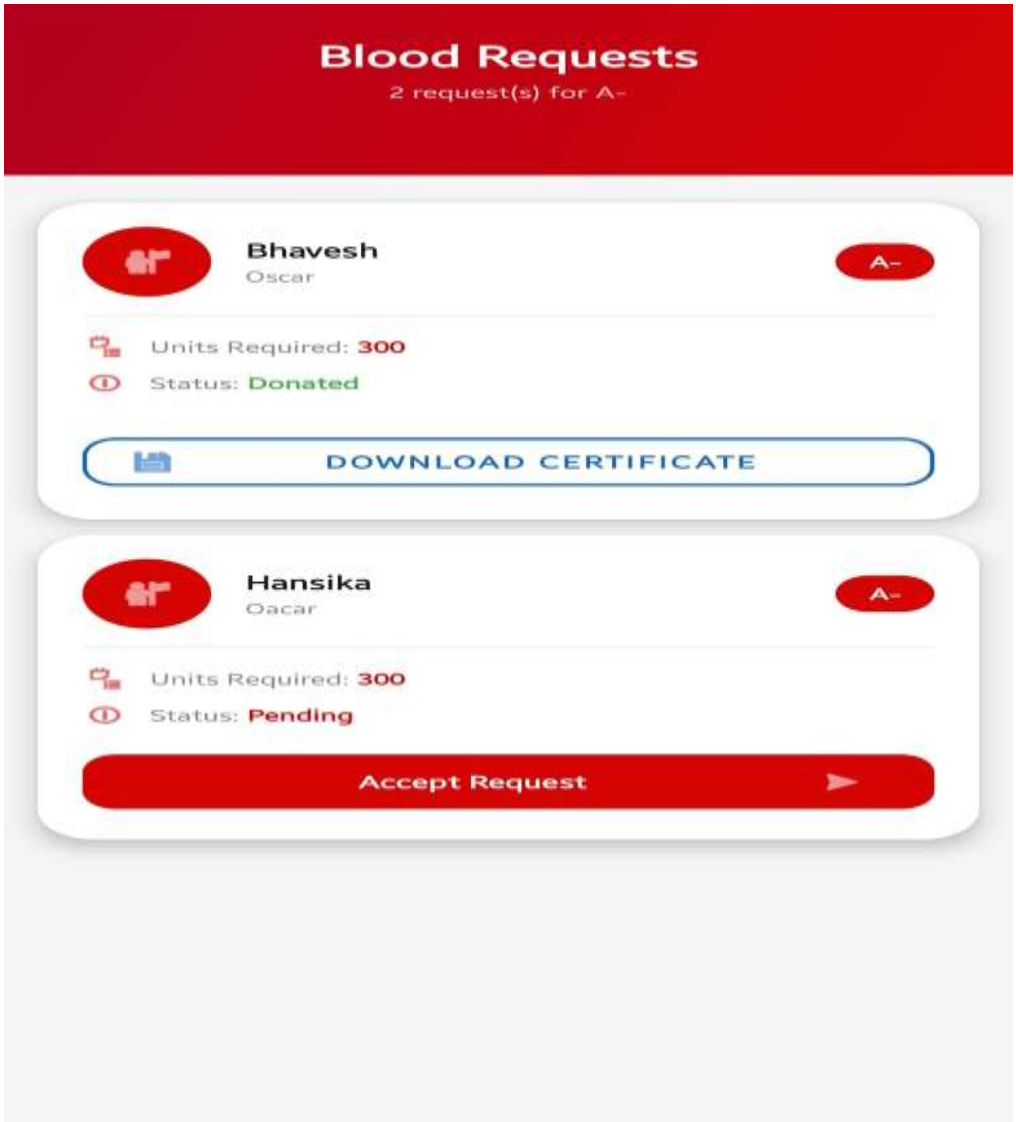

Figure7.8:Blood request interface

This screen displays a dedicated list of blood requests specifically filtered for blood type A-, clearly indicating the total count of pending needs. It presents the requests in a mobile-friendly format, detailing specific requirements such as the recipient's name and the required blood units. Each entry shows the current status, distinguishing between completed donations marked as "Donated" and those still awaiting action marked as "Pending." For completed requests, there is an option to "Download Certificate," adding a layer of recognition for the donor's contribution. Conversely, pending requests feature a prominent red "Accept Request" button, enabling users to quickly commit to a donation. Overall, the design prioritizes clarity and quick action, facilitating a seamless connection between donors and specific blood requirements.

Chapter 8

Conclusion

The Blooddrop application was developed as a native Android platform using Kotlin and Android Studio, with Firebase integrated as the backend service to handle real-time data storage, user authentication, and notifications. The project follows a modular Gradle build structure with AndroidX support, ensuring compatibility across a wide range of Android devices. By connecting donors, recipients, and administrators through a single unified platform, Blooddrop addresses a critical gap in the blood donation ecosystem — making the process of finding and requesting blood faster, more transparent, and more accessible than traditional methods.

The system successfully implements three distinct user roles, each with clearly defined responsibilities. Donors can register, manage their availability, and respond to blood requests, while recipients can post urgent requests and search for compatible donors in their vicinity. The Admin panel provides oversight across the entire platform — including the ability to monitor all active blood requests, manage user accounts, and ban or remove users who violate platform guidelines — ensuring the integrity and trustworthiness of the application's user base.

Looking ahead, Blooddrop has significant potential for further enhancement. Features such as real-time location-based donor matching, in-app chat between donors and recipients, blood camp event scheduling, and integration with hospital management systems could greatly increase the application's impact. The foundation laid by this project — a clean Firebase-backed Android architecture with role-based access control — provides a strong and scalable base for these future improvements. Ultimately, Blooddrop demonstrates how mobile technology can be leveraged to save lives by bridging the gap between those who need blood and those willing to donate it.

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